SPURVANCE

Internship Program

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Project Lab Report

Windows Firewall Configuration

Objective:

To configure and review Windows Firewall inbound rules for SSH (TCP port 22), HTTP (TCP port 80), and HTTPS (TCP port 443), and understand how firewall rules control incoming network connections.

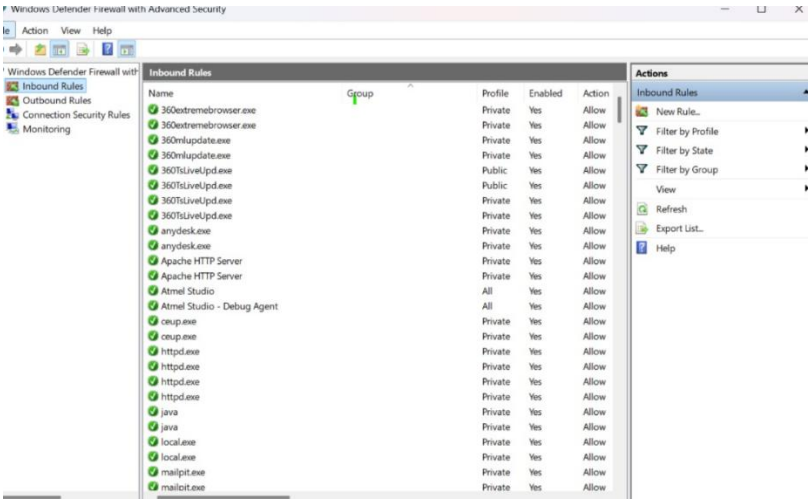
Configuration:

Three inbound rules were configured in Windows Firewall with Advanced Security:

- **Lab - Allow SSH TCP 22:** Allows incoming TCP connections on port 22.
- **Lab - Allow HTTP TCP 80:** Allows incoming TCP connections on port 80.
- **Lab - Allow HTTPS TCP 443:** Allows incoming TCP connections on port 443.

Screenshot 01: Windows Firewall with Advanced Security

This screenshot shows the Windows Firewall with Advanced Security console opened through wf.msc. It provides access to inbound and outbound firewall rules and allows the user to configure network traffic filtering.



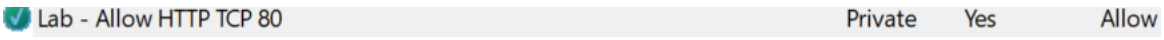
Screenshot 02: SSH Inbound Rule TCP Port 22

This screenshot shows the inbound firewall rule named **Lab - Allow SSH TCP 22**. The rule is enabled, allows TCP connections on port 22, and applies to the Private k network profile. SSH is commonly used for secure remote administration



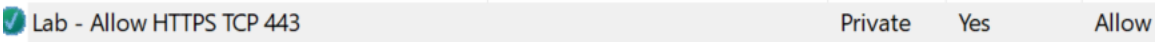
Screenshot 03: HTTP Inbound Rule TCP Port 80

This screenshot shows the inbound firewall rule named **Lab - Allow HTTP TCP 80**. The rule is enabled and allows incoming TCP traffic on port 80 for the Private network profile. HTTP is commonly used for web traffic without transport encryption.



Screenshot 04: HTTPS Inbound Rule TCP Port 443

This screenshot shows the inbound firewall rule named **Lab - Allow HTTPS TCP 443**. The rule is enabled and allows incoming TCP traffic on port 443 for the Private network profile. HTTPS uses TLS to protect web communications.



Screenshot 05: Port 80 Connectivity Test

This screenshot shows the result of running Test-NetConnection localhost -Port 80 in Windows Command Prompt. The output reports TcpTestSucceeded: False, meaning a TCP connection to localhost on port 80 could not be established. This result alone does not confirm that the firewall blocked the connection because a service may not be listening on the port.

```
WARNING: TCP connect to (::1 : 80) failed
WARNING: TCP connect to (127.0.0.1 : 80) failed


ComputerName            : localhost
RemoteAddress           : ::1
RemotePort              : 80
InterfaceAlias          : Loopback Pseudo-Interface 1
SourceAddress           : ::1
PingSucceeded           : True
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded        : False
```

Screenshot 06: Network Port Status Using Netstat

This screenshot shows the output of the netstat -an command, which displays listening ports and active network connections. The output helps review network activity on the computer but does not independently prove that all other ports are blocked by the firewall.

```
netstat -an
```

```
-Command "Test-NetConnection localhost -Port 80"
```

```
Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:902            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:912            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:5357           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49666          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49667          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49668          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49696          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49700          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49701          0.0.0.0:0              LISTENING
  TCP    127.0.0.1:1434         0.0.0.0:0              LISTENING
  TCP    127.0.0.1:6789         0.0.0.0:0              LISTENING
  TCP    127.0.0.1:6789         127.0.0.1:49687        ESTABLISHED
  TCP    127.0.0.1:6789         127.0.0.1:49739        ESTABLISHED
  TCP    127.0.0.1:6791         0.0.0.0:0              LISTENING
  TCP    127.0.0.1:6791         127.0.0.1:31417        ESTABLISHED
  TCP    127.0.0.1:21517        127.0.0.1:49673        TIME_WAIT
  TCP    127.0.0.1:21521        127.0.0.1:49673        TIME_WAIT
```

```
TCP    127.0.0.1:21526        127.0.0.1:36085        TIME_WAIT
TCP    127.0.0.1:21529        127.0.0.1:49673        TIME_WAIT
TCP    127.0.0.1:21532        127.0.0.1:49673        TIME_WAIT
TCP    127.0.0.1:23394        127.0.0.1:49673        TIME_WAIT
TCP    127.0.0.1:23397        127.0.0.1:49673        TIME_WAIT
TCP    127.0.0.1:23402        127.0.0.1:49673        TIME_WAIT
TCP    127.0.0.1:23407        127.0.0.1:49673        TIME_WAIT
TCP    127.0.0.1:23414        127.0.0.1:49673        TIME_WAIT
TCP    127.0.0.1:31412        127.0.0.1:49680        ESTABLISHED
TCP    127.0.0.1:31417        127.0.0.1:6791         ESTABLISHED
TCP    127.0.0.1:34380        127.0.0.1:36086        ESTABLISHED
TCP    127.0.0.1:36083        127.0.0.1:49673        TIME_WAIT
TCP    127.0.0.1:36085        0.0.0.0:0              LISTENING
TCP    127.0.0.1:36086        0.0.0.0:0              LISTENING
TCP    127.0.0.1:36086        127.0.0.1:34380        ESTABLISHED
TCP    127.0.0.1:36092        127.0.0.1:49673        TIME_WAIT
TCP    127.0.0.1:36096        127.0.0.1:36097        ESTABLISHED
TCP    127.0.0.1:36097        127.0.0.1:36096        ESTABLISHED
TCP    127.0.0.1:39795        127.0.0.1:49673        TIME_WAIT
TCP    127.0.0.1:39797        127.0.0.1:49673        TIME_WAIT
TCP    127.0.0.1:49673        0.0.0.0:0              LISTENING
TCP    127.0.0.1:49673        127.0.0.1:21518        TIME_WAIT
TCP    127.0.0.1:49673        127.0.0.1:23394        TIME_WAIT
TCP    127.0.0.1:49673        127.0.0.1:23397        TIME_WAIT
TCP    127.0.0.1:49673        127.0.0.1:23407        TIME_WAIT
TCP    127.0.0.1:49673        127.0.0.1:23414        TIME_WAIT
TCP    127.0.0.1:49673        127.0.0.1:36077        TIME_WAIT
TCP    127.0.0.1:49673        127.0.0.1:36092        TIME_WAIT
TCP    127.0.0.1:49680        0.0.0.0:0              LISTENING
```

```
TCP    127.0.0.1:49687        127.0.0.1:6789         ESTABLISHED
TCP    127.0.0.1:49725        0.0.0.0:0              LISTENING
TCP    127.0.0.1:49739        127.0.0.1:6789         ESTABLISHED
TCP    192.168.0.104:139      0.0.0.0:0              LISTENING
TCP    192.168.0.104:1144     150.171.28.10:443      ESTABLISHED
TCP    192.168.0.104:1943     150.171.109.163:443    ESTABLISHED
TCP    192.168.0.104:1979     172.253.143.188:5228   ESTABLISHED
TCP    192.168.0.104:3864     20.140.48.70:443       ESTABLISHED
TCP    192.168.0.104:6636     172.64.148.235:443     ESTABLISHED
TCP    192.168.0.104:10990    4.213.25.240:443       ESTABLISHED
TCP    192.168.0.104:11009    52.110.18.201:443      ESTABLISHED
TCP    192.168.0.104:11010    52.110.18.201:443      ESTABLISHED
TCP    192.168.0.104:11011    52.110.18.201:443      ESTABLISHED
TCP    192.168.0.104:11018    52.110.18.193:443      ESTABLISHED
TCP    192.168.0.104:11019    52.110.18.193:443      ESTABLISHED
TCP    192.168.0.104:11020    52.110.18.193:443      ESTABLISHED
TCP    192.168.0.104:11021    52.110.18.193:443      ESTABLISHED
TCP    192.168.0.104:15808    110.93.229.253:443     ESTABLISHED
TCP    192.168.0.104:16155    142.250.187.46:443     ESTABLISHED
TCP    192.168.0.104:16574    35.190.80.1:443        SYN_SENT
TCP    192.168.0.104:21423    157.240.227.60:443     ESTABLISHED
TCP    192.168.0.104:23391    104.212.68.83:443      ESTABLISHED
TCP    192.168.0.104:28982    35.190.80.1:443        SYN_SENT
TCP    192.168.0.104:29373    74.125.71.84:443       ESTABLISHED
TCP    192.168.0.104:29583    23.4.84.135:443        CLOSE_WAIT
TCP    192.168.0.104:30202    204.79.197.222:443     ESTABLISHED
TCP    192.168.0.104:34425    172.64.148.235:443     ESTABLISHED
TCP    192.168.0.104:35398    23.215.7.8:443         CLOSE_WAIT
TCP    192.168.0.104:36074    147.182.211.77:443     ESTABLISHED
```

```
TCP    192.168.0.104:36080    10.0.2.15:1515         SYN_SENT
TCP    192.168.0.104:37818    131.253.33.254:443     ESTABLISHED
TCP    192.168.0.104:39591    172.64.148.235:443     ESTABLISHED
TCP    192.168.0.104:39794    159.223.101.159:443    ESTABLISHED
TCP    192.168.0.104:42429    150.171.28.12:443      ESTABLISHED
TCP    192.168.0.104:49228    99.86.18.81:443        ESTABLISHED
TCP    192.168.0.104:50012    150.171.28.11:443      ESTABLISHED
TCP    192.168.0.104:51446    110.93.229.240:443     ESTABLISHED
TCP    192.168.0.104:52341    172.64.155.209:443     ESTABLISHED
TCP    192.168.0.104:52655    172.64.155.209:443     ESTABLISHED
TCP    192.168.0.104:52972    157.240.227.60:443     ESTABLISHED
TCP    192.168.0.104:53589    150.171.27.10:443      ESTABLISHED
TCP    192.168.0.104:58174    13.107.138.254:443     ESTABLISHED
TCP    192.168.0.104:58794    150.171.85.254:443     ESTABLISHED
TCP    192.168.0.104:62980    110.93.229.240:443     ESTABLISHED
TCP    192.168.56.1:139       0.0.0.0:0              LISTENING
TCP    192.168.230.1:139      0.0.0.0:0              LISTENING
TCP    192.168.244.1:139      0.0.0.0:0              LISTENING
TCP    [::]:135               [::]:0                 LISTENING
TCP    [::]:445               [::]:0                 LISTENING
TCP    [::]:5357              [::]:0                 LISTENING
TCP    [::]:49664             [::]:0                 LISTENING
TCP    [::]:49665             [::]:0                 LISTENING
TCP    [::]:49666             [::]:0                 LISTENING
TCP    [::]:49667             [::]:0                 LISTENING
TCP    [::]:49668             [::]:0                 LISTENING
TCP    [::]:49696             [::]:0                 LISTENING
TCP    [::]:49700             [::]:0                 LISTENING
TCP    [::]:49701             [::]:0                 LISTENING
```

```
TCP    [::1]:1434             [::]:0                 LISTENING
TCP    [::1]:42050            [::]:0                 LISTENING
TCP    [::1]:49669            [::]:0                 LISTENING
UDP    0.0.0.0:123            *:*
UDP    0.0.0.0:500            *:*
UDP    0.0.0.0:3702           *:*
UDP    0.0.0.0:3702           *:*
UDP    0.0.0.0:4500           *:*
UDP    0.0.0.0:5050           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5353           *:*
UDP    0.0.0.0:5355           *:*
UDP    0.0.0.0:57196          157.240.227.60:443
UDP    0.0.0.0:57353          *:*
```

```
UDP    0.0.0.0:60578          23.215.7.8:443
UDP    0.0.0.0:62654          142.251.37.74:443
UDP    0.0.0.0:63176          142.251.153.119:443
UDP    127.0.0.1:1900         *:*
UDP    127.0.0.1:50092        *:*
UDP    127.0.0.1:65398        127.0.0.1:65398
UDP    192.168.0.104:137      *:*
UDP    192.168.0.104:138      *:*
UDP    192.168.0.104:1900     *:*
UDP    192.168.0.104:2177     *:*
UDP    192.168.0.104:50091    *:*
UDP    192.168.56.1:137       *:*
UDP    192.168.56.1:138       *:*
UDP    192.168.56.1:1900      *:*
UDP    192.168.56.1:2177      *:*
UDP    192.168.56.1:50088     *:*
UDP    192.168.230.1:137      *:*
UDP    192.168.230.1:138      *:*
UDP    192.168.230.1:1900     *:*
UDP    192.168.230.1:2177     *:*
UDP    192.168.230.1:50089    *:*
UDP    192.168.244.1:137      *:*
UDP    192.168.244.1:138      *:*
UDP    192.168.244.1:1900     *:*
UDP    192.168.244.1:2177     *:*
UDP    192.168.244.1:50090    *:*
UDP    [::]:123               *:*
UDP    [::]:500               *:*
UDP    [::]:3702              *:*
```

```
UDP    [::]:3702              *:*
UDP    [::]:4500              *:*
UDP    [::]:5353              *:*
UDP    [::]:5353              *:*
UDP    [::]:5353              *:*
UDP    [::]:5353              *:*
UDP    [::]:5353              *:*
UDP    [::]:5353              *:*
UDP    [::]:5353              *:*
UDP    [::]:5353              *:*
UDP    [::]:5353              *:*
UDP    [::]:5355              *:*
UDP    [::]:57354             *:*
UDP    [::1]:1900             *:*
UDP    [::1]:50087            *:*
UDP    [fe80::1eef:3349:a638:cf99%14]:1900  *:*
UDP    [fe80::1eef:3349:a638:cf99%14]:2177  *:*
UDP    [fe80::1eef:3349:a638:cf99%14]:50086  *:*
UDP    [fe80::759a:633:a45f:3a93%13]:1900  *:*
UDP    [fe80::759a:633:a45f:3a93%13]:2177  *:*
UDP    [fe80::759a:633:a45f:3a93%13]:50084  *:*
UDP    [fe80::858a:e47e:94ac:507b%11]:1900  *:*
UDP    [fe80::858a:e47e:94ac:507b%11]:2177  *:*
UDP    [fe80::858a:e47e:94ac:507b%11]:50085  *:*
UDP    [fe80::f2b4:8f55:21dc:8092%19]:1900  *:*
UDP    [fe80::f2b4:8f55:21dc:8092%19]:2177  *:*
UDP    [fe80::f2b4:8f55:21dc:8092%19]:50083  *:*
```

Screenshot 07: Final Inbound Firewall Rules

This screenshot shows the three configured inbound rules for SSH (TCP 22), HTTP (TCP 80), and HTTPS (TCP 443). All three rules are enabled, set to Allow, and restricted to the Private network profile.

Lab - Allow HTTP TCP 80	Private	Yes	Allow
Lab - Allow HTTPS TCP 443	Private	Yes	Allow
Lab - Allow SSH TCP 22	Private	Yes	Allow

Conclusion:

In this lab, three inbound rules were configured in Windows Firewall to allow TCP traffic on ports 22 (SSH), 80 (HTTP), and 443 (HTTPS) for the Private network profile. The netstat -an command was used to review listening ports and active network connections. A connectivity test to localhost on port 80 returned TcpTestSucceeded: False, indicating that the connection could not be established. This result does not independently confirm firewall blocking, as no service may have been listening on the port. Further testing is required to verify connectivity and confirm whether other ports are blocked. Overall, this lab provided practical experience in configuring Windows Firewall rules and reviewing network connection status.